

Experimental Networked Embedded Mini Drone - Part I. Consideration of Faults

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Abstract — This paper deals with a networked embedded system. The studied prototype is a 4 rotors mini-drone (helicopter) equipped with an attitude central for system positioning, made of 3 rate gyros, 3 accelerometers and 3 magnetometers. A mathematical model of the drone has been integrated within an observer (for attitude estimation) and a control law is implemented for flight control, thus forming a closed loop system. Its fault free behavior is used as a reference for further analyses. Several possible network technologies are compared. The influence of several types of faults is simulated, including sensor, actuator or network induced faults.

Keywords — Fault Diagnosis, Networked Control System, Embedded Systems, 4 rotors helicopter.

I. INTRODUCTION

MORE and more physical systems are networked controlled (NCS). This may be justified either by their large scale or by the fact that they are embedded. Among these embedded systems, automobile or aeronautic industries are good examples. Network control has many advantages over other techniques: reduced wiring, simpler connections, decreased weight and power, etc.... The signals issued from the sensors and actuators and the controller data need to share the network. NCS study emerges as an interdisciplinary research area combining network and control theory [1].

In particular, diagnostic studies must take into account two points [2] [3]. The first one is that the network itself is a component of the system and thus its failures, malfunctions or simply its specific characteristics must be considered. The possible failures due to the network are very particular, compared to the classical failures such as sensor or actuator failures that Fault Detection and Isolation (FDI) studies have

taken into account until now. The second point is that some phenomena due to network overloading, such as packet losses, transmission delays or data desynchronization, ask for new design of classical FDI algorithms [4] [5]. The first point is particularly emphasized in this paper.

An experimental system is used to illustrate this work. It consists of a mini helicopter, whose 4 blades are actuated by 4 DC motors. The drone is also equipped with an attitude central for system positioning, made of 3 rate gyros, 3 accelerometers and 3 magnetometers. A mathematical model of the drone has been integrated within an observer (for attitude estimation) and a control law is implemented for flight control, thus forming a closed loop system. The importance of having fault tolerance for such a system is evident. Any fault could result in severe performance degradation, or even in a crash.

From this model, some sensor and actuator faults have been studied. In order to take into account the network, two aspects have been tested in simulation. The first one is the simple integration of cascades of faults in the information coming from the models of one or several sensors; this can simulate the frame losses and allows studying the consequences of these losses on the control and state estimation performances. In this work, the cascade of faults has been sent randomly, without focusing specifically on an existing network or actual protocol. The second aspect deals with the integration of the model of a network in the model of the drone. Here, the purpose is to take into account the model of a real network for the drone simulation, since this network will be eventually implemented in the physical experiment. The final aim of the study is to test control and diagnostic strategies for NCS. The drone platform has been chosen because of its fast dynamics and strong real time constraints.

The paper is organized as follows. Section II details the system and its dynamic model. The state observer and the control loop of the system are briefly described. Simulations without network constitute the reference for the subsequent analysis. Section III presents some simulation studies of the main faults related to sensors and actuators. Section IV proposes a preliminary analysis of the faults induced by the network.

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II. PRESENTATION OF THE SYSTEM

A drone with 4 rotors is regarded as a composition of two PVTOL (Planar Vertical Take off Landing) problems whose axes are orthogonal allowing a movement of six degrees of freedom [6]. The four electric motors, driving the four blades, provide the thrust for the movement of the drone (Fig. 1). To develop the dynamic equations, two frames are considered, an inertial frame (or reference) R (position) and a frame B attached to the drone (orientation). The dynamic model is described by the fundamental equations of dynamics, linking accelerations and torques. The orientation (or attitude) of the drone is given by three rotation angles with respect to frame B : yaw (ψ), pitch (θ) and roll (ϕ). ω is the angular velocity of the drone measured by the rate gyros in frame B . The drone attitude is represented by a quaternion q which is a 4 dimensions vector [7]. Modeling the attitude using quaternion has several advantages. For instance, it avoids singularities known as gimbal lock while the dynamical equations are not more intricate.



Fig. 1. The drone helicopter benchmark.

The estimation of attitude, position and velocity is the pre requisite for flight control. For this purpose, a central navigation unit is embedded. It consists of a tri-axes accelerometer $a(a_1, a_2, a_3)$, a tri-axes magnetometer $m(m_1, m_2, m_3)$ and a tri-axes rate gyrometer $g(g_1, g_2, g_3)$. These sensors use MEMS (Micro Electronic and Mechanical Systems) technology to reduce the weight, dimension and cost. Unfortunately, they are subject to bias and drift. With the 9 data provided by the sensors, an estimation of the quaternion $\hat{q}$ is computed together with $\hat{\dot{q}}$ and $\hat{\omega}$. These estimates are provided to the control algorithm (Fig. 2). The reference corresponds to a quaternion and for simplification, it is chosen equal to $q_{ref} = (1 \ 0 \ 0 \ 0)^T$ in the following simulations (except if another value is stated). An extended Kalman filter is used as an observer to obtain the results presented in the next sections [8].

III. INFLUENCE OF SENSOR AND ACTUATOR FAULTS

MEMS sensors used in the attitude central not only present intrinsic uncertainties (for example a bias exists naturally in rate gyros) but may also have several types of faults, such as bias, gain modification, or even the total loss of the information delivered by the sensor. High level functionalities such as the detection of aberrant data or faults in sensors must be added to the existing applications. The same applies to the actuators implemented: the malfunctioning of one of them must be detected as soon as possible to guarantee the system safety.

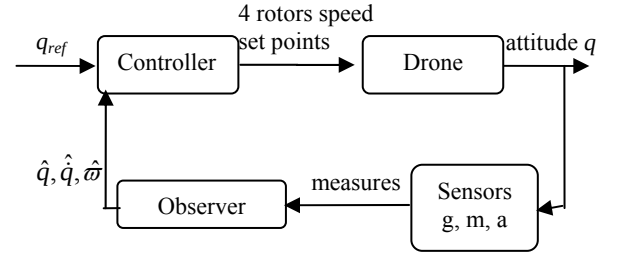


Fig. 2. The control loop.

Before any study of diagnostic strategies, it is highly suitable to envisage the various most common faults that are likely to be observed in the system, from an engineering point of view. Zhang and Jiang [10] have proposed an exhaustive study of vertical or rate gyros faults used for flight control systems. Here, the study is extended to all sensors and actuators. The closed loop system in Fig. 2 is simulated using Matlab-Simulink.

Table 1 Fault scenarios

Figure	Fault pattern	Physical justification
3	Magneto gain modification	scaling error or de-calibration of the sensor
4	Backlash on one motor	loose in efficiency of speed reducers: backlash is always related with the functioning of gears
5	all sensors lost	packet loss

A simulation study of various physical component faults is proposed in this section. It provides useful information for fault detection or controller accommodation. Some representative simulations are provided in Fig. 3 to 5 (see also Table 1). For all simulations except the one given in Fig. 5, the drone is supposed to be at some initial position and the closed loop brings it to the equilibrium $q=(1 \ 0 \ 0 \ 0)^T$ corresponding to $\phi = \psi = \theta = 0$. In the case depicted in Fig. 5, at time $t = 0s$ the drone has an initial attitude and tries to reach the reference $\phi = 10^\circ, \theta = 20^\circ, \psi = 30^\circ$. Then, at time $t = 5s$, the reference is switched to $\phi = \psi = \theta = 0$. At time $t = 10s$, the reference is again changed to the first set point. In this simulation, all sensors are supposed to be faulty

during 50 ms (corresponding to the total loss of 5 data) in a random way. After $t = 10$ s, no more loss of data occurs. When data are lost, the measurement values are set to 1 in order to provide samples to the observer.

Fig. 3a represents the measurement issued from the x-magnetometer m_1 as a normalized value in $[0,1]$. The reference value (unfaulty) is represented by the blue curve; the faulty sensor behavior is represented by the red curve. The “a” part of Fig. 4 represents the speed of one of the 4 rotors. Fig. 5a depicts the measurements acquired from the magnetometer m . As it can be seen, the sensor loss appears randomly.

For all these faulty situations, the effect of the fault on the helicopter attitude is represented in the “b” part of the figures. The three angles yaw-pitch-roll (ψ , θ , ϕ) are represented rather than the quaternion because their interpretation is more intuitive.

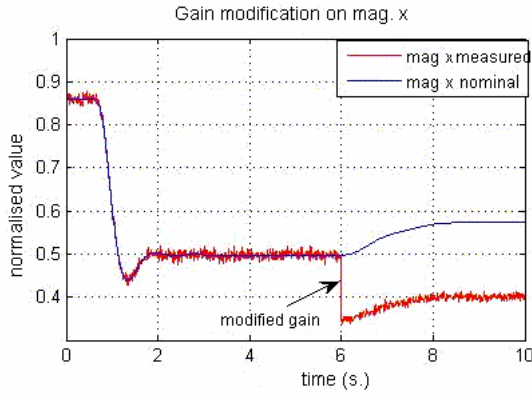


Fig. 3a. Gain modification on the x-magnetometer m_1

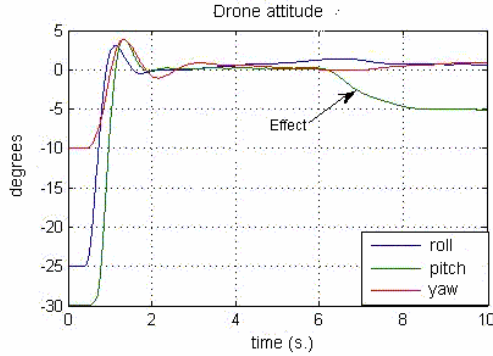


Fig. 3b. Evolution of the drone attitude in the case of a gain modification on the x-magnetometer (yaw (ψ , red), pitch (θ , green) and roll (ϕ , blue))

A change in the gain of magnetometer m_1 (Fig. 3a) induces a constant error on the pitch (Fig. 3b) while a drift on the same sensor induces a pitch drift: this case is not reported in this paper.

Backlash induces small yaw oscillations (Fig. 4b).

As expected, the loss of all sensors induces oscillations of the drone, especially during dynamic phases (Fig. 5b, around $t = 5$ s).

Note that for all these fault scenarios, only the attitude is given. It must be emphasized that a change in the attitude will induce a faulty trajectory of the drone.

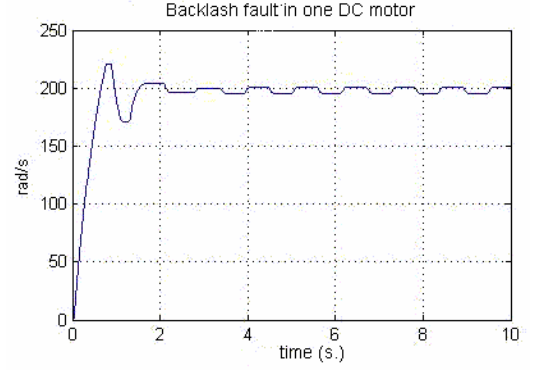


Fig. 4a. Backlash fault in one DC motor (rotor speed)

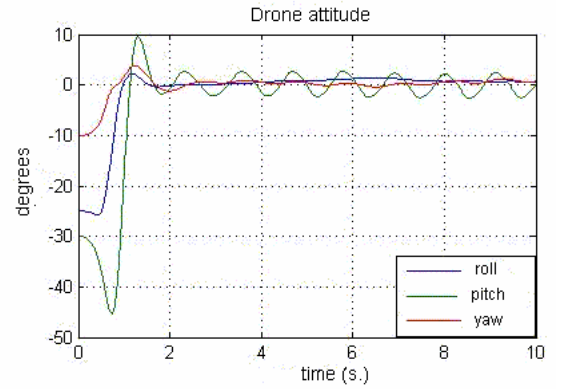


Fig. 4b. Evolution of the drone attitude in the case of a backlash fault in one DC motor (yaw (ψ , red), pitch (θ , green) and roll (ϕ , blue))

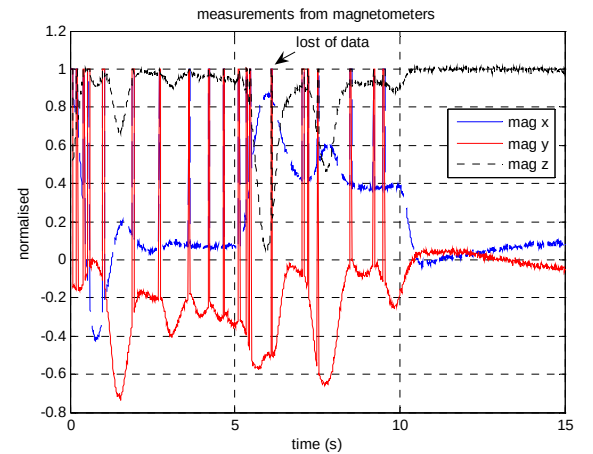


Fig. 5a. Loss of all sensors (magnetometer example).

IV. INFLUENCE OF NETWORK FAULTS

A. Analysis of faults

In this section, the implementation of an embedded network is considered. The real-time constraints as well as the problems induced by the network are taken into account in the case of a wired local network.

The disturbing characteristics induced by the normal functioning of the network are for example delays (random or not, bounded or not), losses of packets or messages, jitters, problems of synchronization, etc. [11] [12] [13] [14]. They may have consequences on the control law or diagnostic strategy. In Fig. 5, the simultaneous loss of sensors can be considered as a packet loss problem. Consequences on the drone stabilization are illustrated in Fig. 5b. Another example of Network fault could be a cut-off of the wire which could be interpreted as a change in the sensor gain (which becomes equal to 0), see Fig. 3.

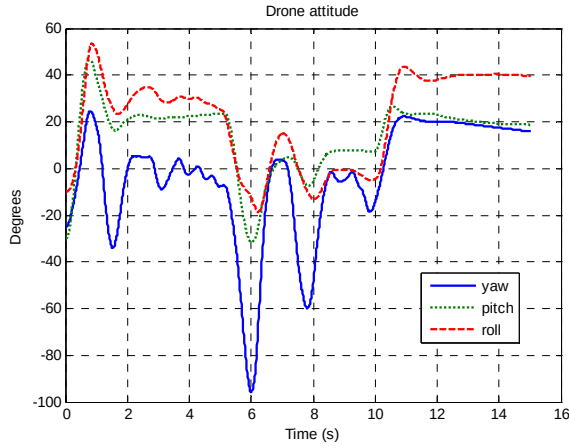


Fig. 5b. Drone attitude when all sensors are lost.

B. TrueTime toolbox application

The purpose is to integrate the model of the network within the fault-free model of the drone (Fig. 6). This can be considered as a hybrid problematic since the model of the drone is either continuous-time or discrete-time whereas the model of the network is event-driven. Thus, it will be possible to compare the attitude control of the drone and its FDI with and without embedded network. Moreover, various networked architectures (wired or wireless networks, various types of networks...) can be simulated.

The system is modeled by using classical Simulink blocks whereas the TrueTime library provides specific blocks for the network interface modeling. This library is developed in C++ language. All the developed files are compiled in Matlab by using an external C++ compiler. TrueTime provides few types of networks which can be used for simulation of the NCS (Ethernet, CAN, Round Robin, TDMA, FDMA, Switched Ethernet + WLAN and ZigBee Wireless networks).

In the architecture of the networked control system, the controller communicates with sensors and actuators via the

network.

The network interconnects 13 nodes: 9 sensors and 4 actuators (Fig. 2). Each node is represented by the TrueTime kernel which encapsulates the D/A and A/D converters, the sender and receiver, etc.

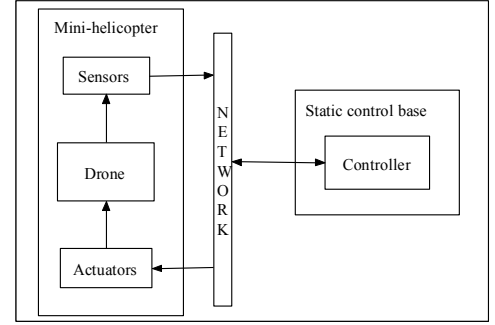


Fig. 6. Structure of the networked control system.

C. CAN and Ethernet networks

In the provided simulation, CAN and Ethernet network models are simulated [15] [16]. These networks are chosen to show the differences in quality of the control between a kind of network which is well adapted for control applications and another type developed for wide fields of applications. Both networks are simulated under the same conditions (data rate, minimum frame size, number of nodes, etc.).

CAN network uses CSMA/AMP arbitration (Carrier Sense Multiple Access with Arbitration on Message Priority) [17]. If the network is busy, the sender waits until the network is free. If a collision occurs (two transmissions are being started within 1 microsecond), the message with the highest priority will continue its transmission.

Nodes connected to Ethernet network use CSMA/CD arbitration (Carrier Sense Multiple Access with Collision Detection). When the network is busy with data transmissions, another node can start sending data but after the first transmission is finished (the message has completely reached its destination). If a collision occurs (two messages are being started within 1 microsecond) then the second sender will back off after a random time. Then the node will try to send the data again. This aspect is also dependent on the data rate of the Ethernet network and on the larger distances between two extremities of the network, thus inducing a minimal length for the frames.

D. Simulations and discussion

A first simulation of the system corresponds to the system without any network to provide a reference behavior. Two other simulations are realized with two different networks, that is a CAN network and an Ethernet network. For the CAN network, a rate of 1 Mbit/s is used. Fig. 7 displays the quaternion representing the drone attitude. In this case, the behavior is almost similar to the reference behavior, except some negligible differences, and a small delay. Fig. 8 shows

an enlargement of Fig. 7, in the case of the component q_2 of the quaternion: slight differences can be observed, that are the consequences of the different characteristics of the network (this point is not emphasized in this paper).

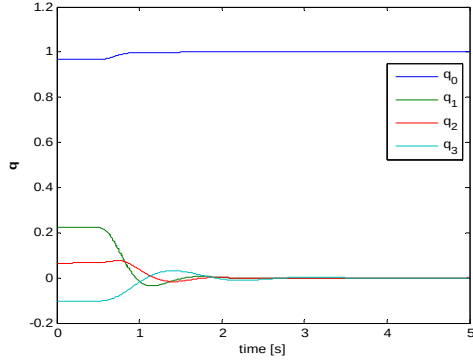


Fig. 7. CAN network with a 1 Mbit/s-data rate.

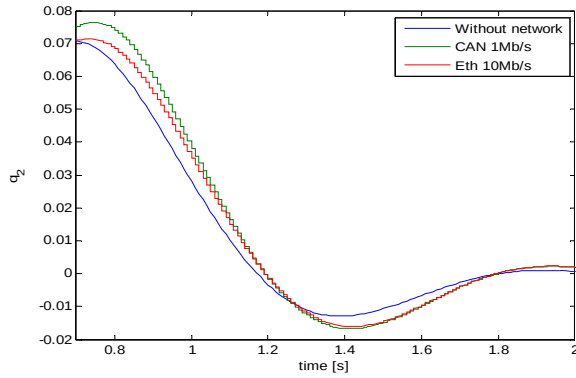


Fig. 8. Comparison of the curves for the quaternion q_2 .

For the Ethernet network, a rate of 10-Mbit/s is chosen at this stage because the simulation results with a lower bit rate (see Fig. 9 where a rate of 1Mbit/s is chosen) without any scheduling policy show that the attitude reference cannot be reached. Thus, the system suffers permanent errors.

The simulations that have been performed show that in the case of the CAN network, the obtained results are as good as in the case of the simulations without network. This result confirms some characteristics of the CAN network, designed for control and avoiding collisions, at least for high-priority messages. Actually, when a collision occurs, the sent data is not interrupted but the node with the highest priority continues its data transmission.

Fig. 9 shows that, the simulation result of the Ethernet network is insufficient when the network data rate is 1 Mbit/s. The reason for this insufficient result can be found in the medium access method (arbitration system). The occurrence of the collisions on the network depends on:

1. some characteristics of the NCS such as the number of nodes connected to the network;
2. the size of the frames or messages;
3. the network data rate;
4. the arbitration strategy;
5. the synchronization among the nodes;

6. the sampling period, etc [18].

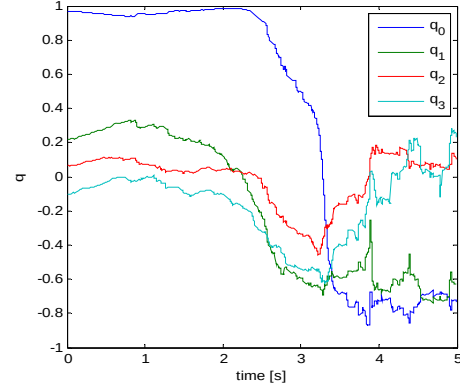


Fig. 9. Ethernet network with a 1 Mbit/s-data rate

The probability of collisions increases if the above characteristics are not correctly tuned. When the mentioned aspects such as the data rate, the number of nodes, the sampling period are not optimally tuned, the system can be destabilized [19] as in Fig. 9. It can be seen that a low bit rate joined with a small sample period of each node represents an NCS with not optimally tuned parameters. Design and tuning of the parameters of the NCS are considered for example in [19].

For instance, when many nodes with short sampling periods are connected to the network with low data rates, this non-equiponderant combination results in the inability of the connected nodes to send the required data within the sampling period. Therefore, the required information (sampled and computed data) is received with a too long delay. It is well known that the arbitration access method of the Ethernet network description can cause delays when a collision occurs. Actually, when this collision is detected, each collision node waits for a random time interval. When the number of collisions is increasing, the delays are also augmenting. The solution to this problem is to increase the data rate: in the present application, it has been necessary to increase the data rate up to 10 Mbit/s.

On the contrary, CAN network is designed for transferring small amount of data in small time intervals [19], and in combination with arbitration with a message priority. Thus the information transfer is ensured without collision. Even if this network works with a lower bit rate, its performance can be compared to the one of the Ethernet network. Therefore, the medium access method in combination with other parameters of the network (sampling period, bit rate, etc.) has significant influence on the embedded system behavior [19].

For the networked control system presented in this paper (Fig. 6), all the connected nodes are supposed to be synchronized. This synchronization increases the probability of collision because each node sends sampled or computed data with the same sampling period, and therefore, at the same time than the other nodes. Thus, when the system is

synchronized, the probability of collisions and the delays (and losses) are rapidly increasing. For the present application, the sampling period of each node is 0.01s which may result in the above mentioned problems. When an Ethernet network with a data rate of 1 Mbit/s is considered, all the conditions for an increase of the collision rate and of the delays (low bit rate, a lot of nodes, small sampling period, etc.) are met. As a consequence, the embedded system could be destabilized. A possible solution here is to schedule the tasks in order to desynchronize the various tasks and relative information and so, to avoid collisions and optimize the available network flow rate. This could tend to the model of a deterministic network.

The simulation results confirm that classical Ethernet networks (that is, without additional mechanisms) are more suitable for applications where there are not a lot of data to be sent into a too small temporal window, with strong temporal constraints. Hence the constraints that are dictated by networked control systems cannot be fulfilled by Ethernet network, at least for the mini-helicopter application, as far as NCS are essentially considered as time-driven systems. This aspect should be taken into consideration when a NCS has to be designed.

V. CONCLUSION

This paper presents a study of various fault modes that can occur in a drone system: sensor or actuator faults and networked induced faults. The simulation tools that are implemented allow modeling at the same time the system and the network behavior. Different fault scenarios for sensor and actuator faults are considered, and their consequences on the system behavior are analyzed. Moreover, some of these faults are reinterpreted in terms of disturbing characteristics induced by the network.

Different kinds of networks with different flow rates and protocol strategies are tested. The results show that for the mini helicopter application, some of them cannot be implemented on the real system because of the malfunctioning of the NCS they will induce. The various configurations that have been simulated show that other parameters (such as rate, arbitration strategy...) also influence the performance characteristics of the networked control systems.

Note that faults on the instrument and controller digital interfaces should also be modeled, for instance as transient faults.

This preliminary work, completed in simulation, gives information about the different fault scenarios that will be implemented on the prototype which is under development. Different distributed FDI strategies and FTC techniques will also be tested in simulation on the real platform.

The future work will consist in studying the adaptation of these algorithms to the network (delays, losses of packets, use of dating) and verifying the robustness of FDI and FTC algorithms, when they are used with a network. The use of a

wireless network will be envisaged, particularly for the communication between the drone and ground stations.

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